



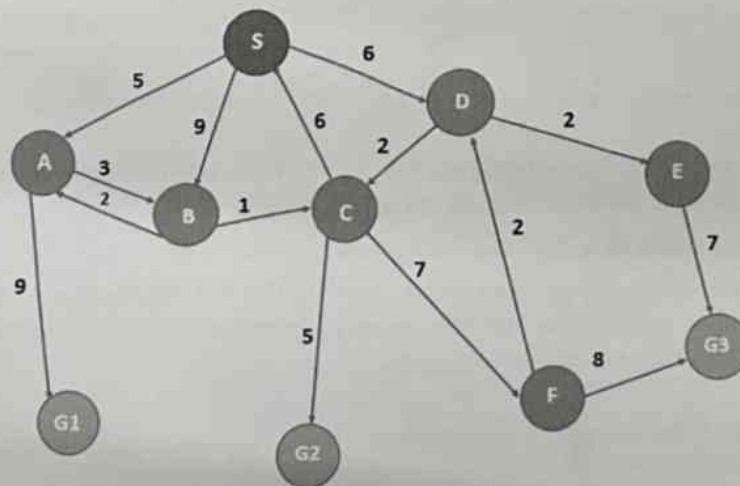
Kishoreganj University
3rd Year 1st Semester B. Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3109: Artificial Intelligence (3 Credits)

Time: 3 Hours

Full Marks: 70

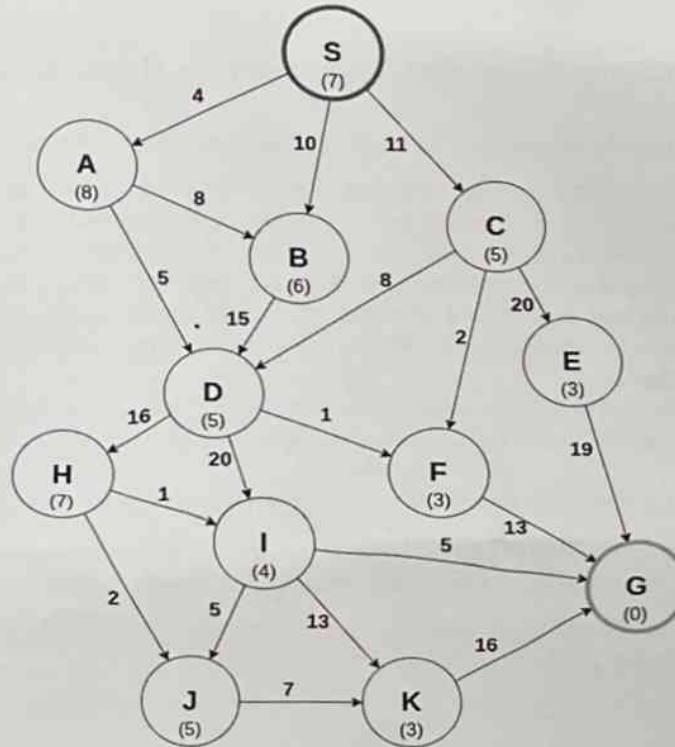
Figures shown in the right margin indicate full marks.
Answer 05 out of 07 questions.

- 1 a. What is Artificial Intelligence (AI)? Do you think it can be both rational and ethical at the same time? 2
- b. Shortly compare the approaches "Acting Humanly" and "Acting Rationally." 2
- c. A smart delivery drone operates in a busy city environment. It must deliver packages efficiently while avoiding collisions, saving battery power, and maintaining customer satisfaction. The environment includes unpredictable weather, moving obstacles, and varying delivery priorities. The drone uses onboard sensors and data from a cloud system to make decisions autonomously. 10
- i. Identify and describe the characteristics of the drone's environment (observable/partially observable, deterministic/stochastic, static/dynamic, discrete/continuous, etc.) with justification.
- ii. Which type of agent is more appropriate for this drone scenario? Justify your answer.
- iii. Draw and label the architecture of the agent that is suitable for the scenario.
- 2 a. Why planning is important in AI? Differentiate between searching and planning. 4
- b. What is blind search? Between blind (uninformed) and informed search strategies, which would you prefer to use in solving AI problems? Justify your answer with reasons. 4
- c. Find the least cost path from source S to one of the goals (G1, G2, G3) using uniform cost search 6

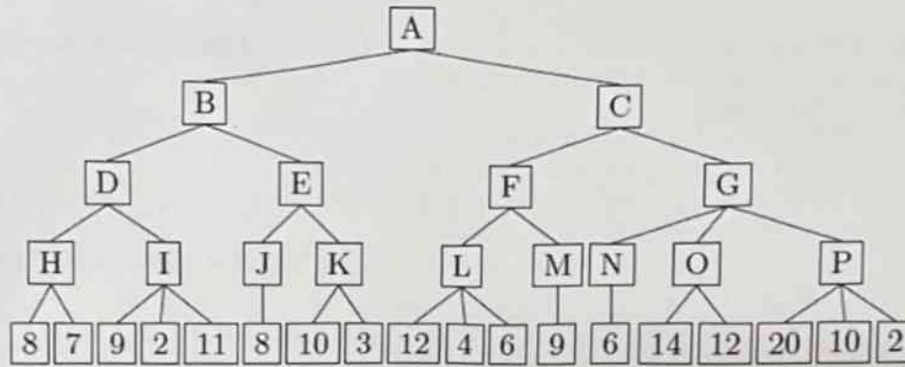


- 3 a. Describe how Artificial Intelligence, Machine Learning, and Deep Learning are related. Explain their hierarchical relationship and illustrate it with examples or a simple diagram to show how each subset fits within the other. 4
- b. Describe the difference between supervised, unsupervised, and reinforcement learning with suitable examples. 4
- c. How artificial neuron is related to biological neuron? Describe the internal structure of an artificial neuron with proper diagram. 6

- 4 a. What is the limitation of depth first search? How it can be solved? 3
- b. Consider the following directed graph. Node S is the source and node G is the destination. 6
The values beside the edges represent the distances between nodes, while the values inside each node (enclosed in parentheses) represent the heuristic estimates of the straight-line distance from that node to the destination. Apply A* search in the graph to find optimal path from S to G.



- c. What is the drawback of hill climbing algorithm? How it can be solved? 5
- 5 a. Why is it important to study Game Theory in Artificial Intelligence? Explain its relevance and applications in AI decision-making and strategic interactions. 4
- b. What do you mean by a zero-sum game? Give an example. 2
- c. Consider the following tree-like graph. Suppose that the top level is a MAX node (the A node is MAX node). Use Minimax algorithm with alpha beta pruning to determine what action A should take and the associated utility. 8



- 6 a. From the given dataset shown in Table 6(a), we train a Naïve Bayes classifier. Each row refers to an apple instance with three categorical features (size, color, and shape) and one class label indicating whether the apple is good or not. 7

Table 6(a): Training Data for Naïve Bayes classifier

RID	Size	Color	Shape	Class: good apple
1	Small	Green	Irregular	No
2	Large	Red	Irregular	Yes
3	Large	Red	Circle	Yes
4	Large	Green	Circle	No
5	Large	Green	Irregular	No
6	Small	Red	Circle	Yes
7	Large	Green	Irregular	No
8	Small	Red	Irregular	No
9	Small	Green	Circle	No
10	Large	Red	Circle	Yes

Given a new apple with features $x = (\text{Small, Red, Circle})$, what is the estimation of $p(y = w_0 | x)$? Would the Naïve Bayes classifier predict $y = \text{yes}$ or $y = \text{no}$ for the apple x ?

- b. The following Figure 6(b) represents a multilayer neural network with one hidden layer. 7

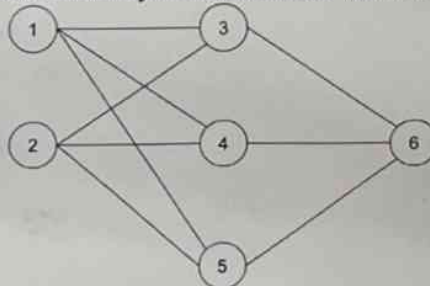


Figure 6(b)

A weight on the connection between nodes i and j is denoted by w_{ij} , such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the weight in the network.

$w_{13} = 0.9$	$w_{36} = 0.3$
$w_{14} = 0.2$	$w_{46} = 0.6$
$w_{15} = 0.1$	$w_{56} = 0.8$
$w_{23} = 0.3$	
$w_{24} = 0.8$	
$w_{25} = 0.5$	

Determine the modified weights for w_{13} and w_{46} , after one iteration of training back propagation algorithm.

- 7 a. Define heuristics and the properties of heuristics in terms of AI-based search algorithms. Describe the characteristics of the following heuristic algorithms: 4
- Iterative Deepening A* (ID-A*)
 - Recursive Best First Search (RBFS)
 - Simplified Memory Bounded A* (SM-A*)
- b. Illustrate a possible solution if an AI agent fails to operate in an environment because of a large number of states and huge amount of operations to perform. 5
- Based on the five components of problem-solving, formulate the problem described in Figure 7(b) (apply the concept of state mapping problem).

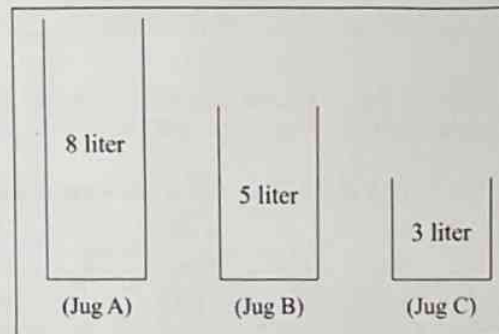


Figure-7(b): A three Jars problem

Problem descriptions:

- There are three jugs A, B, and C with capacities of 8 liters, 5 liters, and 3 liters respectively.
 - Only the 8-liter jug is filled up with water.
 - There are no measuring scales.
 - There is no extra water supply.
 - There are no extra jugs to use.
 - Goal: By a series of pouring back and forth among the three jugs, divide the 8 liters into two equal amounts (4 liters in Jug A and 4 liters in Jug B).
- c. Formulate a state mapping tree based on the problem description in Question 7(b) (Figure-7(b)). 5